POLAR WINDOW

A lifecycle planner for CLPS lunar deliveries, from launch to return,
with the uncertainty and source of every number

Submission report for the 2026 challenge *CLPS Lunar Mission Browser*. All numbers were measured on commit `f421298` of branch `featured`.

1 Introduction

NASA's Commercial Lunar Payload Services (CLPS) programme buys the delivery of science payloads to the Moon from commercial providers [1]. Four CLPS landers have flown: Peregrine Mission One, IM-1, Blue Ghost Mission 1 and IM-2. Their reconstructed trajectories are public in NASA's SPICE archive [2]. The 2026 NASA Space Apps challenge *CLPS Lunar Mission Browser* asks for a tool that lets planners, educators and the public compare landing sites and dates, with the Sun and Earth shown against the local horizon [3].

The first version of POLAR WINDOW answered one question. At a south-polar site and over chosen dates, it reported how long the Sun and Earth stay above the site's own skyline. A delivery raises many more questions. A project manager must know when the vehicle can leave the pad and whether the weather rules allow it. The trip passes through sunlight, shadow and the radiation belts. After landing, the team needs the hours in which a ground station can hear the lander. The launch vehicle's booster comes back to a landing zone on land or at sea. Each payload must also be built, tested and delivered on time. Each answer comes from a different tool and data source, and each tool states its assumptions in its own way, if at all.

This report describes the extension of POLAR WINDOW into a planner that follows one CLPS delivery through six sections: *Ready*, *Launch*, *Trip*, *Surface*, *Contact* and *Return*. The work makes four contributions.

1. A single contract for every number. Each value states its provenance (computed, published, entered by a named person, or assumed), a margin (a ±, a range or a multiplicative factor) or the reason it has none, and the age of any live input. No section returns a verdict, a readiness percentage or a combined score.
2. Trajectory, launch, radiation and entry engines checked against flown missions. The references are NASA's reconstructions of the four CLPS trips and the measured landing of the OSIRIS-REx sample capsule.
3. A process design that separates network access, heavy computation and request handling. Every stored answer records the inputs and engine versions it came from.
4. A disclosure of every planned acceptance tolerance that the engines do not meet, with the measured value beside it.

The evaluation is bounded by public data. Only four CLPS trips have reconstructions, and none of the planned missions publishes a trajectory, a radio link budget or a return state. The engines are therefore checked on the flown cases, and the planned missions receive designed answers labelled as such. Two external references named in the plan could not be run: the General Mission Analysis Tool (GMAT) and the SPENVIS web service. **??** states what replaced each of them.

2 Related work

Illumination and Earth visibility at the lunar poles. Polar lighting has been mapped from Lunar Orbiter Laser Altimeter (LOLA) topography [4] by casting horizons and averaging Sun visibility over years [5]. Later work located persistently illuminated regions in camera images [6] and in higher-resolution terrain models [7]. These studies produce fractions over long periods on a fixed grid. They answer where the light is on average, not what happens at one landing point over one mission's dates. They also report no timing uncertainty per crossing. The surface engine of POLAR WINDOW computes the Sun and Earth from the Navigation and Ancillary Information Facility's SPICE system [8, 9] with the DE440 ephemeris [10]. It reports each duration with a margin derived from the terrain's vertical error.

Trajectory and launch design. High-fidelity mission design is available in open tools such as GMAT, which has a documented verification programme [11]. Its building blocks are standard: Lambert solvers for the initial transfer [12], numerical integration under point-mass and harmonic gravity [13], and lunar gravity fields from GRAIL [14]. These tools require a scripted mission and an analyst. They do not read NASA's CLPS reconstructions, they do not relate a launch day to the lighting rule at the landing site, and they do not label an input as published or assumed. POLAR WINDOW does not compete with them on fidelity. It connects a small, checked subset of their methods to the landing site, the pad and the reconstructions.

Launch weather and forecast verification. Launch weather criteria are published per vehicle as rule sets, for example the Falcon 9 limits on wind at 162 ft and on lightning near the pad [15]. Ensemble systems such as the Global Ensemble Forecast System (GEFS) give a sample of possible weather [16]. Lightning is observed from orbit by the Geostationary Lightning Mapper (GLM) [17, 18], and long wind records come from the MERRA-2 reanalysis [19]. The standard score for a probability forecast is the Brier score [20]. Public presentations of launch weather usually give one chance of a rule being broken. They rarely state the sample behind it or whether that chance has been checked against observations.

Radiation screening. Trapped belt fluxes are commonly taken from the AE8 and AP8 models [21, 22], implemented in the IRBEM library [23] and exposed in Python by SpacePy [24]. Galactic cosmic rays (GCR) can be modelled with one solar modulation parameter [25], and measured doses exist in lunar orbit from CRaTER [26] and on the far-side surface from Chang'e-4 [27]. Large solar particle events are described by published spectra [28]. Engineering tools such as OLTARIS compute dose behind shielding once a trajectory and a geometry are known [29]. They need an account and a finished design, so they do not serve early screening of several candidate trips.

Contact planning. The Deep Space Network (DSN) publishes each antenna's horizon mask and link parameters [30]. Delay-tolerant networks route messages over scheduled contacts with contact graph routing [31]. Neither source combines a station's mask with the terrain around a polar lander, which decides whether Earth is visible at all.

Gap. Each area has mature tools, but each answers one phase with its own conventions. None carries the provenance and uncertainty of a number from the pad to the landing site and back. POLAR WINDOW adds that thread. Its engines are deliberately simpler than the specialist tools above, and every answer names the specialist tool in which it should be confirmed.

References

[1] NASA Science Mission Directorate. CLPS deliveries. NASA Science. URL `https://science.nasa.gov/lunar-science/clps-deliveries/`. Accessed 2026-09-27.

[2] NASA Navigation and Ancillary Information Facility. CLPS SPICE archive (PDS4). Planetary Data System. URL `https://naif.jpl.nasa.gov/pub/naif/pds/pds4/clps/clps_spice/`. Reconstructed trajectories of Peregrine, IM-1, Blue Ghost Mission 1 and IM-2; accessed 2026-09-26.

[3] NASA Space Apps Challenge. CLPS lunar mission browser. Challenge statement, 2026. URL `https://www.spaceappschallenge.org/2026/challenges/clps-lunar-mission-browser/`.

[4] David E. Smith, Maria T. Zuber, Glenn B. Jackson, John F. Cavanaugh, Gregory A. Neumann, Haris Riris, Xiaoli Sun, Ronald S. Zellar, Craig Coltharp, Joseph Connelly, Richard B. Katz, Igor Kleyner, Peter Liiva, Adam Matuszeski, Erwan M. Mazarico, Jan F. McGarry, Anne-Marie Novo-Gradac, Melanie N. Ott, Carlton Peters, Luis A. Ramos-Izquierdo, Lawrence Ramsey, David D. Rowlands, Stephen Schmidt, V. Stanley Scott, George B. Shaw, James C. Smith, Joseph-Paul Swinski, Mark H. Torrence, Glenn Unger, Anthony W. Yu, and Thomas W. Zagwodzki. The lunar orbiter laser altimeter investigation on the lunar reconnaissance orbiter mission. *Space Science Reviews*, 150(1–4):209–241, 2010. doi: 10.1007/s11214-009-9512-y.

[5] E. Mazarico, G.A. Neumann, D.E. Smith, M.T. Zuber, and M.H. Torrence. Illumination conditions of the lunar polar regions using lola topography. *Icarus*, 211(2):1066–1081, 2011. doi: 10.1016/j.icarus.2010.10.030.

[6] Emerson J. Speyerer and Mark S. Robinson. Persistently illuminated regions at the lunar poles: Ideal sites for future exploration. *Icarus*, 222(1):122–136, 2013. doi: 10.1016/j.icarus.2012.10.010.

[7] P. Gläser, F. Scholten, D. De Rosa, R. Marco Figuera, J. Oberst, E. Mazarico, G.A. Neumann, and M.S. Robinson. Illumination conditions at the lunar south pole using high resolution digital terrain models from lola. *Icarus*, 243:78–90, 2014. doi: 10.1016/j.icarus.2014.08.013.

[8] Charles H. Acton. Ancillary data services of nasa's navigation and ancillary information facility. *Planetary and Space Science*, 44(1):65–70, 1996. doi: 10.1016/0032-0633(95)00107-7.

[9] Charles Acton, Nathaniel Bachman, Boris Semenov, and Edward Wright. A look towards the future in the handling of space science mission geometry. *Planetary and Space Science*, 150:9–12, 2018. doi: 10.1016/j.pss.2017.02.013.

[10] Ryan S. Park, William M. Folkner, James G. Williams, and Dale H. Boggs. The jpl planetary and lunar ephemerides de440 and de441. *The Astronomical Journal*, 161(3):105, 2021. doi: 10.3847/1538-3881/abd414.

[11] Steven P. Hughes, Rizwan H. Qureshi, Steven D. Cooley, and Joel J. Parker. Verification and validation of the general mission analysis tool (gmat). In *AIAA/AAS Astrodynamics Specialist Conference*. American Institute of Aeronautics and Astronautics, 2014. doi: 10.2514/6.2014-4151.

[12] Dario Izzo. Revisiting lambert's problem. *Celestial Mechanics and Dynamical Astronomy*, 121(1):1–15, 2015. doi: 10.1007/s10569-014-9587-y.

[13] Oliver Montenbruck and Eberhard Gill. *Satellite Orbits*. Springer Berlin Heidelberg, Berlin, Heidelberg, 2000. doi: 10.1007/978-3-642-58351-3.

[14] Maria T. Zuber, David E. Smith, David H. Lehman, Tom L. Hoffman, Sami W. Asmar, and Michael M. Watkins. Gravity recovery and interior laboratory (grail): Mapping the

lunar interior from crust to core. *Space Science Reviews*, 178(1):3–24, 2013. doi: 10.1007/s11214-012-9952-7.

[15] NASA Kennedy Space Center. Falcon 9 crew dragon launch weather criteria. Technical Report FS-2020-05-568-KSC, National Aeronautics and Space Administration, 2020. URL `https://www.nasa.gov/wp-content/uploads/2021/07/falcon9_crewdragon_launch_weather_criteria_fact_sheet.pdf`.

[16] Hong Guan, Yuejian Zhu, Eric Sinsky, Bing Fu, Wei Li, Xiaqiong Zhou, Xianwu Xue, Dingchen Hou, Jiayi Peng, M. M. Nageswararao, Vijay Tallapragada, Thomas M. Hamill, Jeffrey S. Whitaker, Gary Bates, Philip Pegion, Sherrie Frederick, Matthew Rosencrans, and Arun Kumar. Gefsv12 reforecast dataset for supporting subseasonal and hydrometeorological applications. *Monthly Weather Review*, 150(3):647–665, 2022. doi: 10.1175/mwr-d-21-0245.1.

[17] Steven J. Goodman, Richard J. Blakeslee, William J. Koshak, Douglas Mach, Jeffrey Bailey, Dennis Buechler, Larry Carey, Chris Schultz, Monte Bateman, Eugene McCaul, and Geoffrey Stano. The goes-r geostationary lightning mapper (glm). *Atmospheric Research*, 125–126:34–49, 2013. doi: 10.1016/j.atmosres.2013.01.006.

[18] Katrina Virts, Steven J. Goodman, and Scott Rudlosky. Geostationary lightning mapper (GLM) combined hourly thunder hour. NASA Global Hydrometeorology Resource Center DAAC, 2025.

[19] Ronald Gelaro, Will McCarty, Max J. Suárez, Ricardo Todling, Andrea Molod, Lawrence Takacs, Cynthia A. Randles, Anton Darmenov, Michael G. Bosilovich, Rolf Reichle, Krzysztof Wargan, Lawrence Coy, Richard Cullather, Clara Draper, Santha Akella, Virginie Buchard, Austin Conaty, Arlindo M. da Silva, Wei Gu, Gi-Kong Kim, Randal Koster, Robert Lucchesi, Dagmar Merkova, Jon Eric Nielsen, Gary Partyka, Steven Pawson, William Putman, Michele Rienecker, Siegfried D. Schubert, Meta Sienkiewicz, and Bin Zhao. The modern-era retrospective analysis for research and applications, version 2 (merra-2). *Journal of Climate*, 30(14):5419–5454, 2017. doi: 10.1175/jcli-d-16-0758.1.

[20] GLENN W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950. doi: 10.1175/1520-0493(1950)078<0001:vofeit>2.0.co;2.

[21] James I. Vette. The AE-8 trapped electron model environment. Technical Report NSSDC/WDC-A-R&S 91-24 (NASA-TM-107820), National Space Science Data Center, NASA Goddard Space Flight Center, Greenbelt, MD, 1991. URL `https://ntrs.nasa.gov/citations/19920014985`.

[22] Donald M. Sawyer and James I. Vette. AP-8 trapped proton environment for solar maximum and solar minimum. Technical Report NSSDC/WDC-A-R&S 76-06 (NASA-TM-X-72605), National Space Science Data Center, NASA Goddard Space Flight Center, Greenbelt, MD, 1976. URL `https://ntrs.nasa.gov/citations/19770012039`.

[23] Daniel Boscher, Sebastien Bourdarie, Paul O'Brien, Tim Guild, Daniel Heynderickx, Steve Morley, Adam Kellerman, Christopher Roth, Hugh Evans, Antoine Brunet, Mykhaylo Shumko, Colby Lemon, Seth Claudepierre, Thomas Nilsson, Erwin De Donder, Reiner Friedel, Stu Huston, Kyungguk Min, and Alexander Drozdov. PRBEM/IRBEM: version 5.0.0. Zenodo, 2022.

[24] Jonathan T. Niehof, Steven K. Morley, Daniel T. Welling, and Brian A. Larsen. The spacepy space science package at 12 years. *Frontiers in Astronomy and Space Sciences*, 9:1023612, 2022. doi: 10.3389/fspas.2022.1023612.

[25] Daniel Matthiä, Thomas Berger, Alankrita I. Mrigakshi, and Günther Reitz. A ready-to-use galactic cosmic ray model. *Advances in Space Research*, 51(3):329–338, 2013. doi: 10.1016/j.asr.2012.09.022.

[26] N. A. Schwadron, T. Baker, B. Blake, A. W. Case, J. F. Cooper, M. Golightly, A. Jordan, C. Joyce, J. Kasper, K. Kozarev, J. Mislinski, J. Mazur, A. Posner, O. Rother, S. Smith, H. E. Spence, L. W. Townsend, J. Wilson, and C. Zeitlin. Lunar radiation environment and space weathering from the cosmic ray telescope for the effects of radiation (crater). *Journal of Geophysical Research: Planets*, 117(E12):E00H13, 2012. doi: 10.1029/2011je003978.

[27] Shenyi Zhang, Robert F. Wimmer-Schweingruber, Jia Yu, Chi Wang, Qiang Fu, Yongliao Zou, Yueqiang Sun, Chunqin Wang, Donghui Hou, Stephan I. Böttcher, Sönke Burmeister, Lars Seimetz, Björn Schuster, Violetta Knierim, Guohong Shen, Bin Yuan, Henning Lohf, Jingnan Guo, Zigong Xu, Johan L. Freiherr von Forstner, Shrinivasrao R. Kulkarni, Haitao Xu, Changbin Xue, Jun Li, Zhe Zhang, He Zhang, Thomas Berger, Daniel Matthiä, Christine E. Hellweg, Xufeng Hou, Jinbin Cao, Zhen Chang, Binquan Zhang, Yuesong Chen, Hao Geng, and Zida Quan. First measurements of the radiation dose on the lunar surface. *Science Advances*, 6(39):eaaz1334, 2020. doi: 10.1126/sciadv.aaz1334.

[28] JOSEPH H. King. Solar proton fluences for 1977-1983 space missions. *Journal of Spacecraft and Rockets*, 11(6):401–408, 1974. doi: 10.2514/3.62088.

[29] Robert C. Singleterry, Steve R. Blattnig, Martha S. Clowdsley, Garry D. Qualls, Chris A. Sandridge, Lisa C. Simonsen, Tony C. Slaba, Steven A. Walker, Francis F. Badavi, Jan L. Spangler, Aric R. Aumann, E. Neal Zapp, Robert D. Rutledge, Kerry T. Lee, Ryan B. Norman, and John W. Norbury. Oltaris: On-line tool for the assessment of radiation in space. *Acta Astronautica*, 68(7–8):1086–1097, 2011. doi: 10.1016/j.actaastro.2010.09.022.

[30] Jet Propulsion Laboratory. DSN telecommunications link design handbook. Technical Report 810-005, Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA. URL `https://deepspace.jpl.nasa.gov/dsndocs/810-005/`. Module 301, antenna horizon masks; accessed 2026-09-27.

[31] Giuseppe Araniti, Nikolaos Bezirgiannidis, Edward Birrane, Igor Bisio, Scott Burleigh, Carlo Caini, Marius Feldmann, Mario Marchese, John Segui, and Kiyohisa Suzuki. Contact graph routing in dtn space networks: overview, enhancements and performance. *IEEE Communications Magazine*, 53(3):38–46, 2015. doi: 10.1109/mcom.2015.7060480.